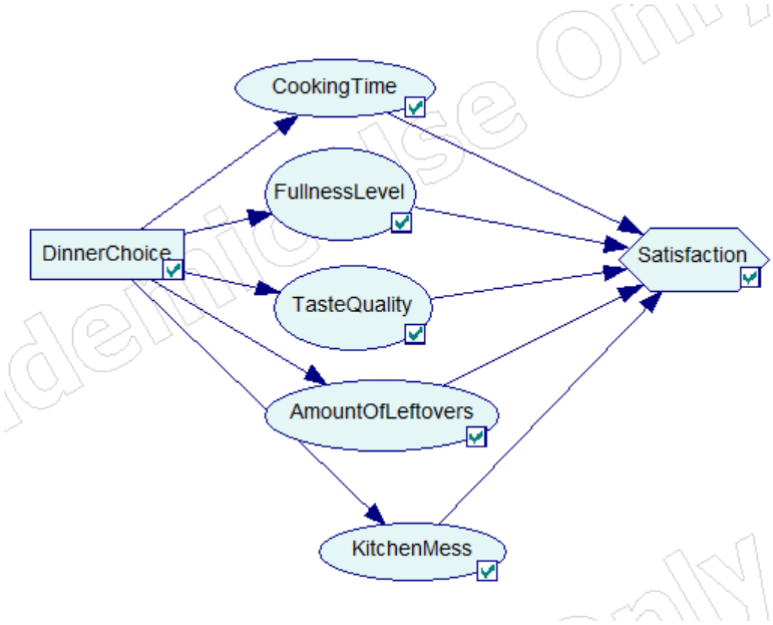
Assignment 5

Decision network

This model addresses the decision of selecting a dinner menu from three options: nachos, pasta, or wok. The problem balances trade-offs between cooking time, fullness, taste, leftovers and mess.



Decision network

Probability distributions

DinnerChoice	Nachos	Pasta	Wok
Fast	0.8	0.2	0.7
Slow	0.2	0.8	0.3

CookingTime

DinnerChoice	Nachos	Pasta	Wok
Perfect	0.4	0.6	0.8
Uncomfortable	0.6	0.4	0.2

FullnessLevel

DinnerChoice	Nachos	Pasta	Wok
Good	0.8	0.7	0.75
Bad	0.2	0.3	0.25

TasteQuality

DinnerChoice	Nachos	Pasta	Wok
None	0.7	0.5	0.6
Some	0.3	0.5	0.4

AmountOfLeftovers

DinnerChoice	Nachos	Pasta	Wok
Low	0.4	0.3	0.6
High	0.6	0.7	0.4

KitchenMess

CookingTime	Fast																Slow															
FullnessLevel	Perfect								Uncomfortable								Perfect								Uncomfortable							
TasteQuality	Good				Bad				Good				Bad				Good				Bad				Good				Bad			
AmountOfLefto...	None		So...		None		So...		None		So...		None		So...		None		So...		None		So...		None		So...		None		So...	
KitchenMess	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High
Value	90	75	1...	85	50	35	60	45	70	55	80	65	30	15	40	25	75	60	85	75	30	20	45	30	55	40	65	50	15	0	25	10

Satisfaction

Assumptions

The model assumes that the uncertain variables, CookingTime, TasteQuality, KitchenMess, AmountOfLeftovers, and FullnessLevel, are conditionally independent given the MealChoice. While in reality a rushed meal might taste worse, this simplification makes the model more manageable. Furthermore, each uncertain variable is restricted to two states to maintain a managable utility table.

Optimal choice

Expected utilities for different policies:

Nachos	61.0352
Pasta	52.98
Wok	**69.536**

Expected utilities

The decision support system evaluated the three meal options based on the principle of maximum expected utility. After running the model, the software calculates an expected utility value for Nachos≈61, Pasta≈53, and Wok≈70.

Thus, the results indicate that **Wok** is the optimal choice.